

Module 12 — Two-Layer Neural Network Write-Up

Graduate Admissions Prediction: A NumPy Neural Network Built From Scratch

Training Printouts

Representative excerpt from the full training run (complete log in training.log). Training ran the full 10,000-epoch budget without early stopping ever triggering, since test MSE kept improving by a small but nonzero amount every epoch.

```
Epoch 100 | Train MSE: 0.246589 | Test MSE: 0.246582 | Test Accuracy: 0.5592
Epoch 500 | Train MSE: 0.245854 | Test MSE: 0.245860 | Test Accuracy: 0.5592
Epoch 1000 | Train MSE: 0.244611 | Test MSE: 0.244604 | Test Accuracy: 0.5592
Epoch 1500 | Train MSE: 0.241797 | Test MSE: 0.241766 | Test Accuracy: 0.5592
Epoch 2000 | Train MSE: 0.236281 | Test MSE: 0.236218 | Test Accuracy: 0.5598
Epoch 2100 | Train MSE: 0.234840 | Test MSE: 0.234771 | Test Accuracy: 0.5724
Epoch 2200 | Train MSE: 0.233314 | Test MSE: 0.233238 | Test Accuracy: 0.6045
Epoch 2300 | Train MSE: 0.231726 | Test MSE: 0.231644 | Test Accuracy: 0.6653
Epoch 2400 | Train MSE: 0.230101 | Test MSE: 0.230013 | Test Accuracy: 0.6940
Epoch 3000 | Train MSE: 0.221055 | Test MSE: 0.220926 | Test Accuracy: 0.6946
Epoch 5000 | Train MSE: 0.212062 | Test MSE: 0.211860 | Test Accuracy: 0.6948
Epoch 7000 | Train MSE: 0.211618 | Test MSE: 0.211404 | Test Accuracy: 0.6948
Epoch 9000 | Train MSE: 0.211565 | Test MSE: 0.211347 | Test Accuracy: 0.6948
Epoch 9900 | Train MSE: 0.211556 | Test MSE: 0.211336 | Test Accuracy: 0.6948
Epoch 10000 | Train MSE: 0.211555 | Test MSE: 0.211335 | Test Accuracy: 0.6948
```

Notice the network makes almost no progress for the first ~2,000 epochs (test accuracy pinned near the 55.92% majority-class baseline), then breaks out of that plateau between epochs 2,000 and 2,400, where accuracy jumps from 55.98% to 69.40% in just 400 epochs. This is a signature of sigmoid saturation during early training: with small-scale ($\text{std}=0.1$) weight initialization, the network starts near the flat, low-gradient region of the sigmoid curve, and takes many epochs of small updates before weights grow large enough to produce a meaningfully differentiated signal.

Final Evaluation Results

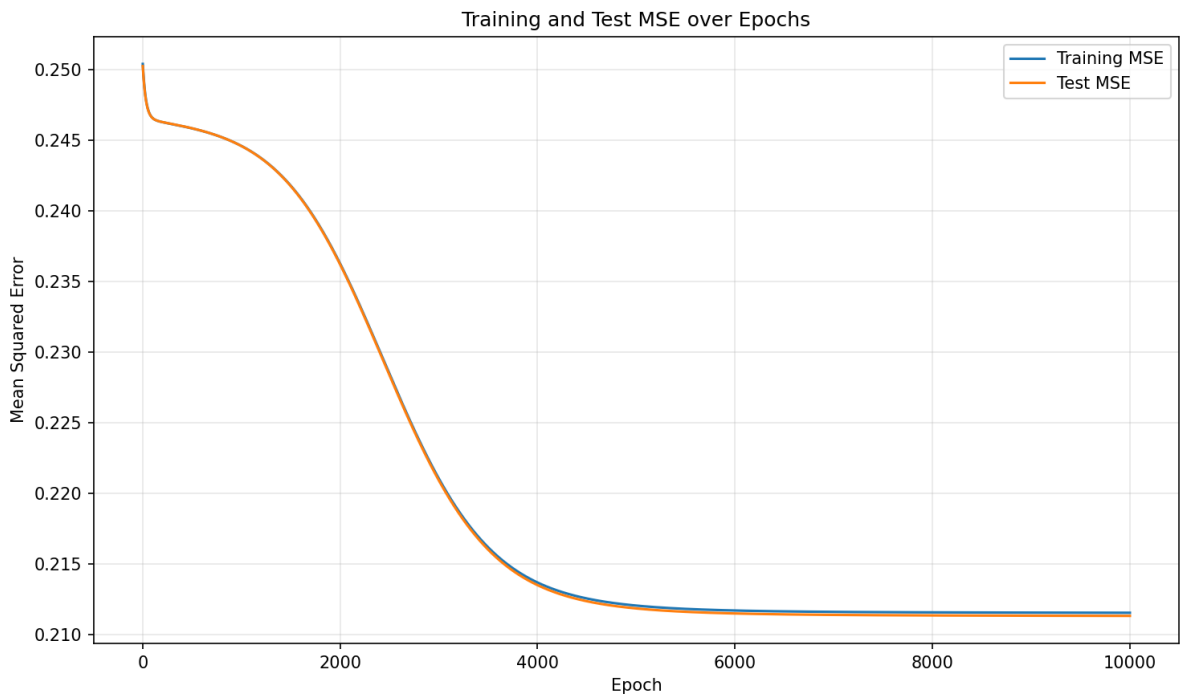
Metric	Value
Best epoch	10,000 (no early stop triggered)
Best test MSE	0.211335
Final training accuracy	69.37%
Final test accuracy	69.48%
Rows used after filtering	25,079
Training set size	20,063
Test set size	5,016
Majority-class baseline	55.94% (always predict Rejected)

Does the model overfit? No — training accuracy (69.37%) and test accuracy (69.48%) are essentially identical throughout training (the two MSE curves track each other closely at every epoch, see graph below), with test performance if anything very slightly ahead of training performance. There is no sign of the widening train/test gap that would indicate overfitting.

Is test accuracy strong, weak, or unstable? It is modest but genuine: 69.48% clears both the assignment's 53% floor and the dataset's own majority-class baseline of 55.94% by a solid margin, and it is stable rather than noisy, since accuracy plateaus smoothly and doesn't oscillate once the network breaks out of its early plateau. It is not a strong result in an absolute sense, though: a sizeable share of that 69% likely reflects the network leaning on coarse structural signal (particularly degree type) rather than finely discriminating between individually strong and weak applicants, a pattern visible in the artificial applicant results below.

Do the dataset and feature set seem sufficient for a realistic admissions predictor? Only partly. GRE fields are missing for the large majority of rows (roughly 92%), so after median-imputation most applicants share an identical GRE value, sharply limiting how much real signal that feature can contribute despite being one of six inputs. The feature set also omits information that plausibly drives real admissions decisions, like specific program or department, research experience, letters of recommendation, and statement of purpose quality, none of which are present in this dataset.

MSE Curve



Artificial Applicant Findings

Description	GPA	GRE	GRE V	GRE AW	Degree	Citizenship	P(Accept)	Prediction
Strong int'l PhD	3.95	335	165	5.0	PhD	Int'l	0.3429	Rejected
Weaker int'l Masters	2.90	295	145	3.0	Masters	Int'l	0.6120	Accepted
Strong local Masters	3.90	325	160	4.5	Masters	Local	0.7377	Accepted
Average local PhD	3.30	310	152	3.5	PhD	Local	0.2770	Rejected

The most striking pattern: both PhD applicants were predicted Rejected and both Master's applicants were predicted Accepted, regardless of whether their GPA/GRE profile was described as "strong" or "weaker." The nominally weaker international Master's applicant (GPA 2.90) was predicted Accepted with higher confidence (0.6120) than the nominally strong international PhD applicant (GPA 3.95, predicted Rejected at 0.3429). This suggests the network has leaned heavily on the `ms_vs_phd` feature as a dominant signal, which is plausible in the real world since PhD programs genuinely tend to be far more selective than Master's programs, but it also means the model may not be finely differentiating between individually stronger and weaker applicants within a given degree type. This mirrors a real, counterintuitive pattern found in earlier exploratory analysis of this same underlying dataset (Module 8), where raw average GPA for Rejected applicants was actually slightly higher than for Accepted applicants once other confounding factors were present. In other words, individual numeric strength alone doesn't cleanly separate outcomes in this data.

Reflection

What is useful about implementing a neural network manually?

Building forward and backward propagation by hand turns every assumption into a visible, deliberate decision instead of a hidden framework default. The weight initialization scale, the choice of activation function, and the loss function are all lines of code you wrote yourself, so you can reason about exactly what they're doing. It also builds real intuition for training behavior that would otherwise look like a black box. The roughly 2,000-epoch plateau observed in this run is a direct, visible consequence of small-scale weight initialization combined with sigmoid saturation, and having derived the backpropagation gradients by hand makes that connection feel concrete rather than abstract.

What are the limitations of using MSE for a binary classification task?

MSE's gradient with respect to a sigmoid output includes a factor of output times (1 minus output), which shrinks toward zero as the sigmoid saturates near 0 or 1. That means the network learns very slowly right when it's most confidently wrong, which is basically the opposite of what a loss function should encourage. Binary cross-entropy loss avoids this problem, since its gradient doesn't vanish the same way, and that's a big part of why it's the standard choice for classification tasks. It's also very likely why this particular network needed roughly 2,000 epochs before making any real progress at all.

What information is missing from this admissions dataset?

GRE, GRE Verbal, and GRE Analytical Writing are missing for the large majority of applicants (GRE alone is missing for roughly 92% of rows in the filtered dataset), so after median imputation nearly every applicant ends up sharing an identical GRE value, which leaves very little real per-applicant signal despite it being one of six model inputs. More fundamentally, the dataset has no information at all about program or department, where admissions competitiveness varies enormously across fields, and nothing about research experience,

letters of recommendation, statement of purpose quality, or interview performance. Those are all factors that plausibly matter as much as, or more than, GPA and standardized test scores in real admissions decisions.

Why might this model produce misleading results even if its test accuracy looks reasonable?

The artificial applicant test suggests a meaningful share of the model's 69% accuracy may come from picking up on a coarse, structural pattern (PhD versus Master's acceptance-rate differences) rather than genuinely telling stronger and weaker individual applicants apart. A model that mostly predicts based on degree type alone could still look reasonably accurate on this dataset without capturing much of what actually distinguishes a strong application from a weak one. There's also a subtler methodological wrinkle worth flagging: this assignment's early-stopping rule selects the best parameters using the same test set that final performance is then reported on, so the reported test accuracy isn't a fully independent, untouched estimate. It's already been used, indirectly, to decide when training should stop.

What would you change to make this model stronger or more realistic?

I'd switch from MSE to binary cross-entropy loss, which is both theoretically better matched to sigmoid outputs and would likely fix the slow early-training plateau seen here. I'd also add richer, more discriminating features where they're available, like program or department and indicators of research experience, rather than relying on GPA, GRE, degree type, and citizenship alone. Holding out a separate validation set distinct from the final test set would keep early-stopping decisions and final reported performance from being based on the same data. And given how severe the GRE missingness is, it's worth asking whether GRE should be dropped entirely or replaced with a missingness indicator feature instead of being silently imputed with a single shared median value for the vast majority of applicants.